Science

Year 4

Student Manual

Prime Books · Science & Lab Year 4 · Student Book

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Before you begin

Unit 1 · Living things

Unit 1 also includes a Final project, Check your progress, Evaluation, What can you do, and a Key words round-up.

Looking ahead

INTRODUCTION

How this book works

Science is a way of asking questions about the world and finding answers you can trust. In this first unit you will look inside your own body, at the 206 bones that hold you up, protect your organs and let you move.

How to help

- **Work together.** Read the page aloud, then let your child lead.
- **Talk lots.** The questions start conversations, they do not test.
- **Keep it fun.** Short, happy sessions beat long, tired ones.

A note for grown-ups

This book follows the spirit of an international primary programme, interpreted afresh for Prime School. Every unit grows new skills out of a topic children already care about, and finishes with something to make, draw or write.

HOW TO USE THIS BOOK

Reading the page

Each section follows the same rhythm, so you always know where you are.

PANEL	WHAT IT DOES
Learning outcomes	What you will be able to do by the end.
Getting started	A warm-up that wakes up what you know.
Key words	The vocabulary of the topic, defined.
Worked example	A model solution, with the reasoning shown.
Questions	Practice, from straightforward to demanding.
Investigation	A hands-on experiment with equipment.
Safety	How to carry out the work safely.
Think like a scientist	A task that builds scientific thinking.
Summary checklist	'I can' statements to check your progress.

How to work

Take your time and look at the pictures: they are there to help you. When you see a new word, say it out loud. In Talking time there are no wrong answers, only good ideas.

GETTING READY

Before Unit 1

You do not need much: a pencil, some crayons, a notebook, and someone to work with.

Your toolkit

- Your eyes, for looking carefully at the pictures.
- Your voice, for reading out loud and sharing ideas.
- Your notebook, for writing and drawing what you find.

A little warm-up

1. What do you already know about this topic?
2. What would you most like to find out?
3. Who could you ask if you got stuck?

These are exactly the things Unit 1 will help you with. When you can answer them, you are ready to begin.

UNIT 1

1

Living things

Bones, skeletons and how you move

IN THIS UNIT, YOU WILL

- ✓ Name some of the bones in the human body.
- ✓ Explain why animals need a skeleton.
- ✓ Describe how bones and muscles work together to move you.
- ✓ Compare different kinds of skeleton.

✦ GETTING STARTED

Press gently on your arm, then on your ear, then on your knee. Which parts feel hard, and which feel bendy? With a partner, list three places on your body where you can feel a bone right under the skin.

UNIT 1 · TOPIC 1.1

Bones and skeletons

Learn

All the bones in your body together make your **skeleton**. An adult human skeleton has **206 bones**. You were born with about 300, but as you grew, some of them joined together, or **fused**. That is why a grown-up has fewer bones than a baby.

Bones are not all the same size. The longest bone in your body is the **femur**, the big bone in your thigh. The smallest is the **stapes**, deep inside your ear, and it is only about three millimetres long, smaller than a grain of rice.

Figure 1.1 The human skeleton, with some bones labelled.

BONE	WHERE IT IS
skull	Around your brain.
ribs	Curved bones around your chest.
spine	Down the middle of your back.
femur	Your thigh: the longest bone.
stapes	Inside your ear: the smallest bone.

QUESTIONS

1. How many bones does an adult human skeleton have?
2. Why does a baby have more bones than a grown-up?
3. Name the longest bone and the smallest bone in your body.
4. Feel for your ribs. How do you think they are arranged, and why?

✧ DID YOU KNOW?

Your bones are alive. They are always being rebuilt, a tiny bit at a time, all through your life. Bones are also surprisingly light: all 206 of them together are only about a seventh of your body weight.

UNIT 1 · TOPIC 1.2

Why we need a skeleton

Learn

A skeleton does three important jobs.

- It gives your body **shape**. Without it you would be a floppy heap.
- It gives your body **support**, holding you up so you can stand and sit.
- It **protects** the soft parts inside you. Your skull protects your brain, and

your ribs protect your heart and lungs.

QUESTIONS

1. Write down the three jobs a skeleton does.
2. Which bone protects your brain?
3. Which bones protect your heart and lungs?
4. Imagine an animal with no skeleton at all. How do you think it would move?

∴ THINK LIKE A SCIENTIST

Making a model. Build a model arm from two strips of card joined with a paper fastener, and add a rubber band as a muscle. Show your model to the class and explain one thing it shows well and one thing it cannot show. Knowing what a model cannot do is just as important as knowing what it can.

UNIT 1 · TOPIC 1.3

Skeletons and movement

Learn

Bones cannot move on their own. They are pulled by **muscles**. Where two bones meet, there is a **joint**, and the joint lets the bones move.

When you bend your arm, a muscle at the front, the **biceps**, gets shorter and pulls the bone up. When you straighten your arm, a muscle at the back pulls it back down. Muscles work in pairs like this, because a muscle can only pull, and never push.

Figure 1.2 A joint at the elbow, where two bones meet.

QUESTIONS

1. What is a joint?
2. Why do muscles have to work in pairs?
3. Name three joints in your body and say what each one lets you do.

KEY WORDS

- **skeleton**: all the bones in a body together.
- **bone**: a hard part of the skeleton.
- **fused**: joined together to make one bone.
- **joint**: the place where two bones meet and can move.
- **muscle**: a part of the body that pulls on bones to move them.

UNIT 1 · TOPIC 1.4

Different kinds of skeletons

Learn

Not all animals have bones inside them. Scientists sort skeletons into three kinds.

KIND	WHERE IT IS	EXAMPLE
internal skeleton	Bones on the inside.	Human, dog, fish, bird
external skeleton	A hard shell on the outside.	Beetle, crab, snail
no skeleton	Soft body, no hard parts.	Earthworm, jellyfish

An animal with an external skeleton has a problem: its shell cannot grow. So it has to shed the old one and grow a new, bigger one. A crab does this several times as it grows.

QUESTIONS

1. Which kind of skeleton does a crab have?
2. Why must a beetle shed its skeleton as it grows?
3. Sort these animals by skeleton kind: cat, snail, earthworm, sardine.

☑ SUMMARY CHECKLIST

- ✓ I can name some bones in the human body.
- ✓ I can explain the three jobs a skeleton does.
- ✓ I can describe how bones, joints and muscles move me.
- ✓ I can compare internal, external and no skeleton.

UNIT 1 · FINAL PROJECT

Build a moving model

★ FINAL PROJECT

Build a moving model

In a group, build a model of a human arm or leg that really moves. Use card for the bones, a paper fastener for the joint and rubber bands for the muscles. Then present it to the class.

What to hand in

- Your working model, with the bones, joint and muscles labelled.
- A short explanation of how the muscles make it move.
- One honest sentence about what your model cannot show.

How it will be judged

Criterion	Weight
Does the model actually move correctly?	40%
Labelling and explanation	30%
Honesty about its limits	20%
Teamwork	10%

UNIT 1 · CHECK YOUR PROGRESS

1. How many bones are in an adult human skeleton?
2. Name the three jobs of a skeleton.
3. Which bone protects the brain?
4. What is a joint, and what does it let bones do?
5. Why can a muscle only pull, and never push?
6. Give one animal with an internal skeleton and one with an external skeleton.
7. Why does an animal with an external skeleton have to shed it?

UNIT 1 · EVALUATION

Be honest. Colour the circle that matches how you feel about each outcome: green means confident, amber means nearly there, red means needs more work.

I CAN …	GREEN	AMBER	RED
name some bones in the body	○	○	○
explain what a skeleton does	○	○	○
describe how I move	○	○	○
compare kinds of skeleton	○	○	○

UNIT 1 · WHAT CAN YOU DO?

✓ WHAT CAN YOU DO?

- [ ] Name some of the bones in the human body.
- [ ] Explain why animals need a skeleton.
- [ ] Describe how bones, joints and muscles work together.
- [ ] Compare internal, external and no skeleton.

UNIT 1 ROUND-UP

skeleton · bone · fused · skull · ribs · spine · femur · stapes · joint · muscle · internal skeleton · external skeleton

GLOSSARY

Unit 1

skeleton: all the bones in a body together.

bone: a hard part of the skeleton.

fused: joined together to make one bone.

joint: the place where two bones meet and can move.

muscle: a part of the body that pulls on bones to move them.

femur: the thigh bone, the longest in the body.

stapes: the smallest bone, inside the ear.

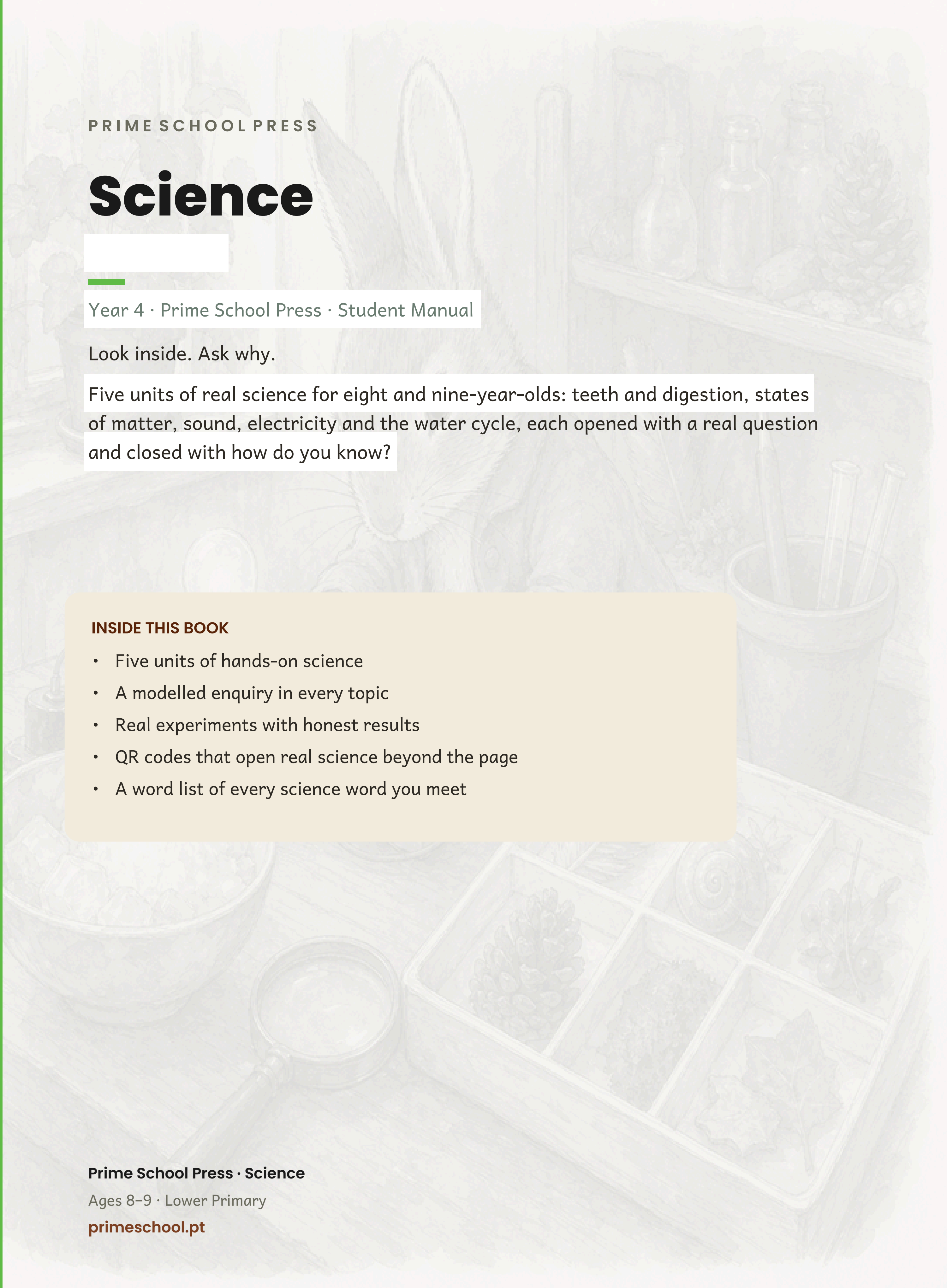
SOURCES & REFERENCES

Where everything came from

All text and diagrams are original to this edition. The factual claims can be checked here. Links were live as of July 2026.

§ UNIT 1 · LIVING THINGS

1. **The human skeleton. An adult skeleton has 206 bones; a baby is born with about 300, which fuse during growth.**
   https://en.wikipedia.org/wiki/List_of_bones_of_the_human_skeleton
2. **The femur and the stapes. The femur is the longest bone and the stapes, in the middle ear, the smallest.**
   https://kidshealth.org/en/kids/bones.html
3. **Bones and body weight. The skeleton makes up roughly 15 per cent of body weight.**
   https://www.healthline.com/health/how-many-bones-does-a-baby-have



PRIME SCHOOL PRESS

Science

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Look inside. Ask why.

Five units of real science for eight and nine-year-olds: teeth and digestion, states of matter, sound, electricity and the water cycle, each opened with a real question and closed with how do you know?

INSIDE THIS BOOK

- Five units of hands-on science
- A modelled enquiry in every topic
- Real experiments with honest results
- QR codes that open real science beyond the page
- A word list of every science word you meet

Prime School Press · Science

Ages 8–9 · Lower Primary

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